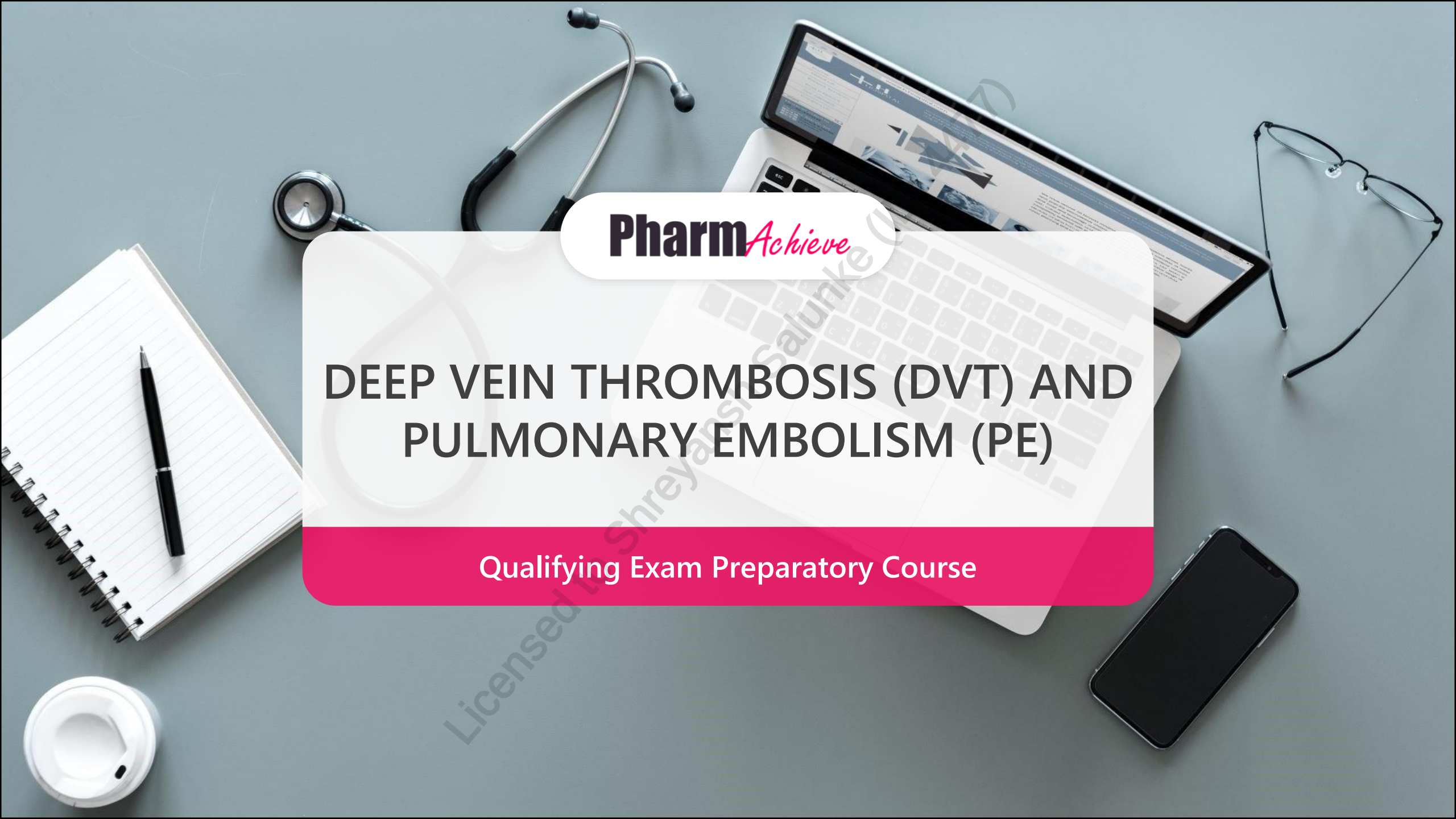




Pharm*Achieve*

DEEP VEIN THROMBOSIS (DVT) AND PULMONARY EMBOLISM (PE)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

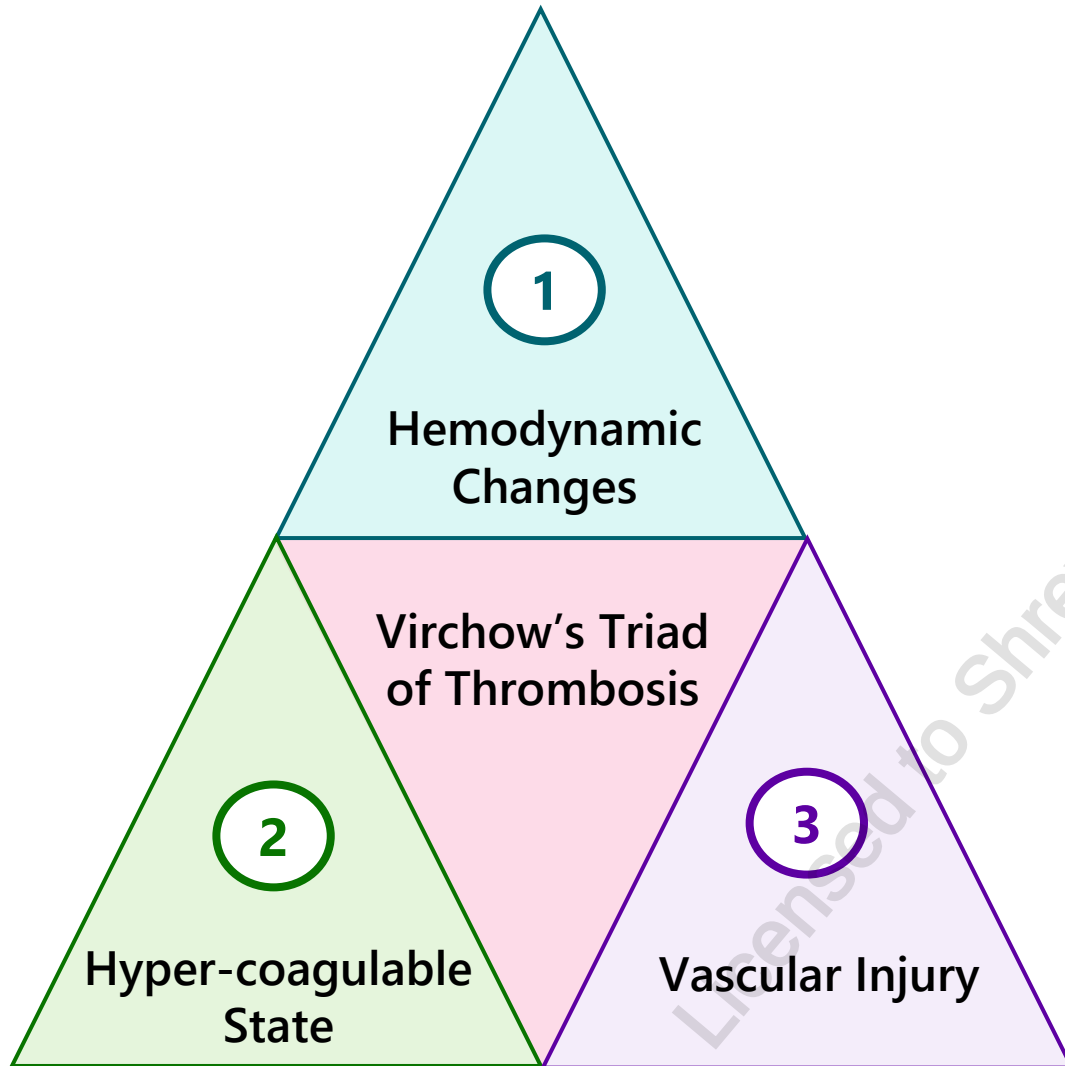
**Leadership &
Stewardship**

Professionalism

LEGEND

AE	Adverse Effect	GI	Gastrointestinal	SCr	Serum Creatinine
APTT	Activated Partial Thromboplastin Time	HIT	Heparin-Induced Thrombocytopenia	SNRI	Serotonin-Norepinephrine Reuptake Inhibitor
ASA	Acetylsalicylic Acid	Hrs	Hours	SOB	Shortness Of Breath
BID	Twice Daily	Hx	History	SSRI	Selective Serotonin Reuptake Inhibitor
C/I	Contraindications	ICU	Intensive Care Unit	Tx	Treatment
CAT	Cancer-Associated Thrombosis	IM	Intramuscular	UFH	Unfractionated Heparin
CBC	Complete Blood Count	INR	International Normalized Ratio	VKA	Vitamin K Antagonist
CHF	Congestive Heart Failure	IPC	Intermittent Pneumatic Compression	VTE	Venous Thromboembolism
CKD	Chronic Kidney Disease	IV	Intravenous	WBC	White Blood Cell
CrCl	Creatinine Clearance	Kg	Kilogram		
CVC	Central Venous Catheter	LMWH	Low Molecular Weight Heparin		
CVD	Cardiovascular Disease	MD	Doctor of Medicine		
CYP	Cytochrome P450	MI	Myocardial Infarction		
D/C	Discontinue	NOACs	Novel Oral Anticoagulants		
D/I	Drug Interaction	NSAID	Non-Steroidal Anti-Inflammatory Drug		
DOAC	Direct Oral Anticoagulant	MOA	Mechanism Of Action		
DOC	Drug of Choice	PE	Pulmonary Embolism		
DVT	Deep Vein Thrombosis	PGP	P-glycoprotein		
ESR	Erythrocyte Sedimentation Rate	PO	Per Os (By Mouth)		
ER	Emergency Room	PTS	Post-Thrombotic Syndrome		
GA	General Anesthetic	Q	Every		
GCS	Graduated Compression Stockings	QHS	At Bedtime		
		SC	Subcutaneous		

PATHOPHYSIOLOGY



Virchow's Triad of Thrombosis

1. Hemodynamic Changes

- Alteration in blood flow
- Blood stasis or turbulence

2. Hypercoagulable State

- Change in clotting functions that may be inherited or acquired
- Fibrin production (coagulation) exceeds fibrin degradation (fibrinolysis)

3. Vascular Injury

- Damage to the blood vessels' endothelium
- Activates the coagulation cascade

PATHOPHYSIOLOGY

Disruption of coagulation system: **inappropriate clot formation**

Types of Blood Clots

Venous thromboembolism (VTE)	Blood clot in a vein	} DVTs and PEs are types of VTEs
Deep vein thrombosis (DVT)	Blood clot formed in the deep veins (usually the leg)	
Pulmonary embolism (PE)	Blood clot that breaks apart and travels to the lung	



Thrombus may remain **asymptomatic**, **lyse**, **cause obstruction**, **propagate**, and/or **embolize**

CLINICAL PRESENTATION

Deep Vein Thrombosis

- Classic symptoms of affected extremity: swelling, pain, erythema
- Important to identify proximal vs. distal DVT (proximal has higher risk of PE)
- Physical exam may show palpable cord on calf OR unilateral edema
- ↑ ESR/WBC + D-Dimer

Pulmonary Embolism

- Wide spectrum of features:
 - Asymptomatic
 - Shock/sudden death
- Most common symptoms:
 - Dyspnea that is followed by: pleuritic pain, cough, syncope, tachypnea
- ↑ ESR/WBC + D-Dimer

RISK FACTORS

Persistent Risks

- Strong family history
- Prior thrombotic event
- Older age (≥ 65 years)
- Male sex
- Active cancer (receiving chemotherapy, recurrent or progressive disease)
- Active inflammatory bowel disease
- Active autoimmune disorders (e.g. rheumatoid arthritis)
- Chronic infections
- Chronic immobility (e.g. spinal cord injury)
- Obesity
- Nephrotic syndrome
- Thrombophilias (Factor V Leiden mutation or protein C or S deficiency)

Transient Risks

- Major surgery with GA ≥ 30 minutes
- Recent hospitalization (≥ 3 days)
- Immobilization or confined to bed due to acute illness (≥ 3 days)
- Trauma with fractures
- Leg injury with reduced mobility (≥ 3 days)
- Pregnancy
- Cesarean section
- Long-haul flight
- Estrogen therapy



A risk factor can now be identified in **80% of patients** with VTEs

DRUG RELATED CAUSES

Estrogen

- Oral contraceptives, estrogen replacement therapy, selective estrogen receptor modulators
- **Estrogen** ↑ **serum clotting factor production**, reduces production of natural anticoagulants (e.g. protein S, antithrombin)
- **Estrogen + smoking** ↑ the risk for VTE significantly

Heparin

- Due to **heparin-induced thrombocytopenia (HIT)**
- Usually caused by UFH, but may be caused by LMWH

Chemotherapy Agents

- Fluorouracil, tamoxifen, busulfan, mitomycin, pegaspargase, cyclophosphamide, methotrexate (high dose), cisplatin, paclitaxel, everolimus, thalidomide, bevacizumab (non-exhaustive list)

HEPARIN-INDUCED THROMBOCYTOPENIA (HIT)

Defined as **low platelet blood** counts due to **heparin therapy** which predisposes patients to thrombosis (**thrombin generation**)

Non-Immune-Mediated (Type 1)

- **Non-immunoglobulin-mediated**
- Frequency: 10-20%
- Mild ↓ of platelet count; rarely $< 100 \times 10^9/L$
- Onset is 1-4 days
- **Observe** – platelet count will return to baseline

Immune-Mediated (Type 2)

- **Immunoglobulin-mediated response to heparin**
- **Platelets fall $>30\%$ vs. baseline**
- **Onset is 5-10 days after start of heparin**
- May develop within 24-48 hours in those who have had recent (3-6 months prior) exposure to heparin
- Treatment:
 - **Cessation of heparin**
 - **Start alternative non-heparin anticoagulant agent:** argatroban, danaparoid, bivalirudin, fondaparinux, rivaroxaban - may transition to warfarin once patient stabilized and platelets $\geq 150 \times 10^9/L$
 - Treat for at least 4 weeks if patient does not have thrombosis and at least 3 months if patient has thrombosis

HEPARIN-INDUCED THROMBOCYTOPENIA (HIT)

Non-heparin anticoagulants for the treatment of HIT

	Fondaparinux*†	Argatroban	Bivalirudin*	Danaparoid	Rivaroxaban*‡
Severe renal impairment (CrCl <30 mL/min)	Avoid	No dose adjustment required	Adjust infusion rate	Adjust dose based on anti-Xa levels	CrCl 15-29 mL/min – no adjustment, use cautiously (off-label) CrCl <15 mL/min – Avoid
Administration	SC	IV	IV	IV	PO
Routine lab monitoring	Not required	aPTT	aPTT	Anti-Xa	Not required

* Not approved for the treatment of HIT

† Rare cases of fondaparinux-induced HIT have been reported

‡ Apixaban and dabigatran have been used to treat patients with HIT, however, there is less data than for rivaroxaban



NOTE: concurrent use of argatroban and warfarin results in falsely elevated INR

VTE COMPLICATIONS

Post-thrombotic syndrome (PTS)	Chronic thromboembolic pulmonary hypertension	Recurrent DVT or PE within 10 years
• Risk increases 2- to 4-fold after proximal DVT • Clot dilates venous valves • Blood remains static even after clot removed due to valve dysfunction • Leads to chronic lower extremity swelling, leg pain, skin ulceration, predisposition to future VTE	• 1-4% of patients after PE	• Approximately one third of patients after DVT

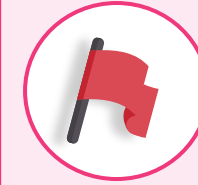
RED FLAGS

Patients should see **MD** if:

- Persistent bleed from cuts or nosebleeds
- Blood in urine or stool
- Increased swelling or pain in affected extremity

Patients to go to **ER** if:

- SOB
- Chest pain
- Black tarry stools
- Severe sudden headache or slurred speech
- Hemoptysis (coughing up blood)



GOALS OF THERAPY

- ✓ Prevent recurrence of DVT/PE
- ✓ Prevent complications of VTE
- ✓ Reduce risk of morbidity and mortality
- ✓ Minimize AEs of treatment

NON-PHARMACOLOGICAL MEASURES - VTE

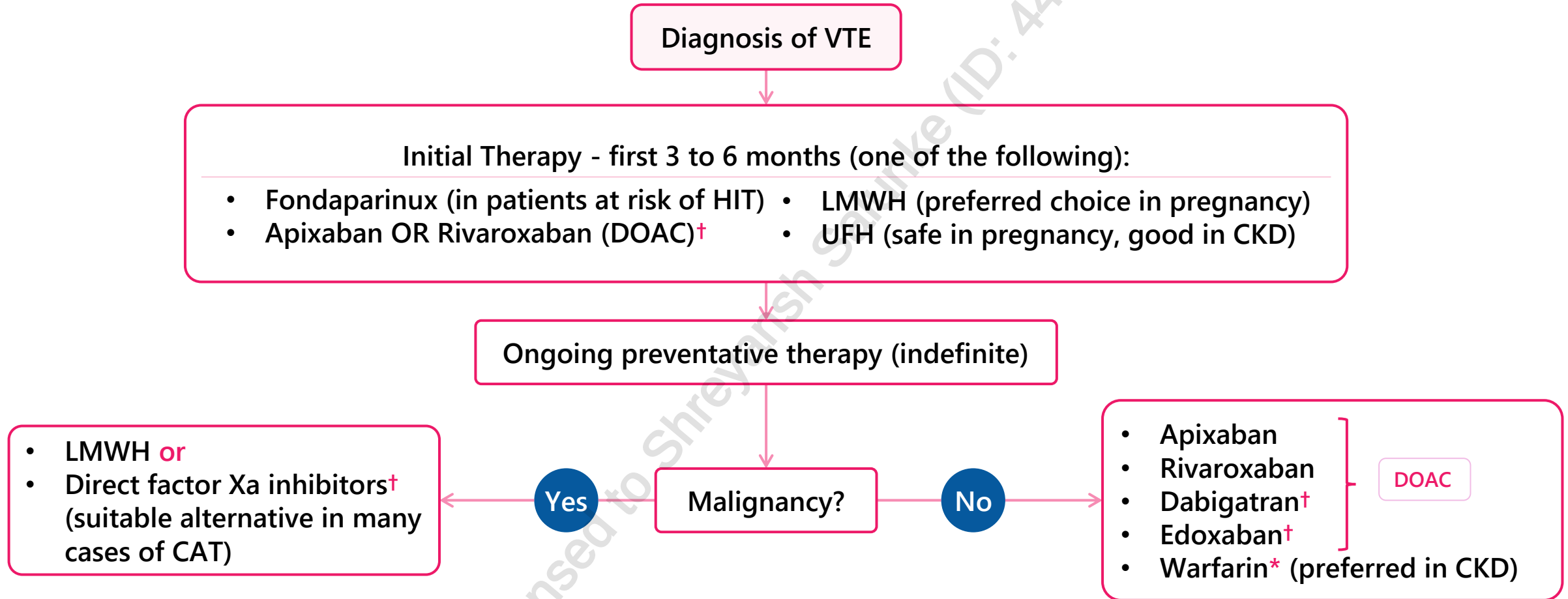
Graduated compression stockings

- (~60% efficacy); applies pressure to the legs to ↑ velocity of lower limb blood flow
- **Do not use for prevention of PTS** – evidence proved no statistically significant difference in placebo versus stocking group for incidence of PTS after DVT
- Can help reduce symptoms (e.g. pain, leg edema) if they are limiting ambulation

Ambulation

- Resolution of pain and swelling may be faster
- Recommended as soon as feasible
- Gradually increase exercise training as tolerated by patient

TREATMENT ALGORITHM - VTE



* Warfarin has a slow onset; start with LMWH / UFH ("bridging": overlap with warfarin for at least 5 days AND until INR is ≥ 2.0 for 2 consecutive days)

[†] 5-10 days of LMWH required before starting edoxaban or dabigatran!

DURATION OF ANTICOAGULATION THERAPY

Categories of VTE	Duration of anticoagulant treatment
- Provoked by transient risk factor (surgery, hospitalization or plaster cast immobilization, all within 3 months; estrogen therapy, pregnancy, prolonged travel (>8 hours), lesser leg injuries or immobilization more recently (within 6 weeks). - The stronger the provoking reversible risk factor e.g., recent major surgery, the lower is the expected risk of recurrence after stopping anticoagulant therapy if the risk factor has resolved)	3 months
First unprovoked VTE (in absence of transient risk factor or active cancer)	Minimum of 3 months then reassess
Proximal unprovoked DVT or PE with no or only minor risk factors for bleeding	Minimum 3 months. Long-term therapy with annual review
Isolated symptomatic distal DVT or isolated distal DVT with risk factors for extension	3 months (patient preference)
Second unprovoked VTE	Minimum of 3 months then reassess for indefinite therapy. For those with no or only minor risk factors for bleeding, long-term therapy is suggested with annual review (indefinite therapy is suggested if moderate risk of bleeding, and 3 months is suggested if there is a high risk of bleeding (patient preference)
Cancer-associated VTE	Minimum of 3-6 months then reassess. Continue if overt evidence of cancer or continuing to receive anti-cancer therapy

PHARMACOLOGICAL MEASURES - HEPARINS

Drug	MOA	Safety	D/I
Unfractionated Heparin (UFH) - Limited by the need for hospitalization, a narrow therapeutic range, lab monitoring, and increased HIT risk	- Activates antithrombin to accelerate the inactivation of factor Xa, and thrombin (IIA). - Prevents growth of clot - Allows the body's own system to degrade the clot	AEs: - Bleeding - HIT (↓risk with LMWH) - Osteoporosis (↓risk with LMWH) - Hypersensitivity reactions	Drugs that ↑ bleeding risk (SSRI, SNRI, ASA, NSAID, alcohol, P2Y12 inhibitors)
Low Molecular Weight Heparin (LMWH) – dalteparin, enoxaparin, tinzaparin	- Same as heparin - Inhibits Xa greater than UFH but less active on antithrombin	C/I: Active bleed, Hx of HIT, severe bleeding risk, recent stroke	

PHARMACOLOGICAL MEASURES - HEPARINS

UFH (continuous IV infusion, SC; not IM as may cause large hematomas)

- Weight-based dosing; safe in renal failure
- Antidote: protamine sulphate 1 mg IV for every 100 units of UFH
- Monitoring: Target aPTT 1.5-2.5 times normal; monitor aPTT Q6H until within therapeutic range
- Limited to patients with severe renal impairment ($\text{CrCl} < 20\text{-}30 \text{ ml/min}$), those at high risk of bleeding who may require rapid reversal, and patients who have very recently received thrombolytic therapy

LMWH (SC)

- Enoxaparin (Lovenox®): 1 mg/kg SC Q12H or 1.5 mg/kg Q24H
- Dalteparin (Fragmin®): 100 units/kg SC Q12H (consider in patients $> 100 \text{ kg}$) or 200 units/kg SC Q24H (preferred)
- Tinzaparin (Innohep®): 175 units/kg SC Q24H
- More predictable anticoagulation dose response than UFH, ↓incidence of HIT, ↓need for routine monitoring, ↓major bleeding risk
- **Antidote: protamine sulphate** (but cannot neutralize LMWH completely)
- Dose adjustment required in renal dysfunction (typically $< 30 \text{ mL/min}$)
- Monotherapy for full duration of Tx in patients with active cancer and those with DVT in pregnancy

PHARMACOLOGICAL MEASURES - WARFARIN

Drug	Safety	D/I
Warfarin (racemic mixture of S and R enantiomers)	AEs: bleeding, bruising, purple toes syndrome Caution: Hx of GI bleed, recent neurosurgery, alcoholic liver disease	Metabolized by CYP 2C9 (S), 1A2 (R), and 3A4 (R) Antifungals, antibiotics, acetaminophen (>1g/day), anti-platelets, anticoagulants, vitamin K rich foods (e.g. leafy greens) , cranberries, alcohol, amiodarone, and fluoxetine

COMMENTS

- Slow onset: Initial anticoagulant effect on INR may be seen as soon as 24-72 hours, however, full therapeutic effect generally seen between 5-7 days after initiation. Start concurrently with LMWH/UFH (**bridging required**)
- **Target INR 2.0-3.0**
- **Reversal agents: vitamin K**, Octaplex® (increase in the levels of vitamin K-dependent coagulation factors), fresh frozen plasma
- **Safe in renal failure**; no dose adjustment required

BRIDGING WARFARIN WITH LMWH FOR VTE TREATMENT

- Initial warfarin treatment must be combined with immediate-acting agent
 - LMWH/UFH for at least 5 days and until INR at least 2.0 for 2 days in a row
- Initial dosing typically 5 mg, but therapeutic dose is highly variable
 - Elderly, frail, and low body weight: consider starting 2-3 mg daily
 - Young, healthy, larger patients: consider initial dose of 7.5-10 mg daily
- Frequent monitoring necessary until stable in-range INR reached
 - INR testing may then be reduced to every 2-4 weeks
 - Changes to concomitant medications and new comorbidities warrants more frequent INR testing
- Patients encouraged to maintain consistent diet
 - **Do not avoid vitamin K-rich foods** – be consistent with weekly intake
 - Low intake of vitamin K can be associated with more unstable INR results

PHARMACOLOGICAL MEASURES - INDIRECT FACTOR XA INHIBITOR

Drug	Safety	D/I
Fondaparinux	• ↓ Bleed risk vs UFH/LMWH • Thrombocytopenia • Hypersensitivity reactions	Drugs that ↑ bleeding risk

COMMENTS

- Treatment: 5 mg SC daily if <50 kg, 7.5 mg SC daily if 50-100 kg, 10 mg SC daily if >100 kg
- **Safe: CrCl 30-50 ml/min; C/I: CrCl < 30 mL/min**
- Measure CBC at baseline and periodically; monitor for signs of bleeding daily
- **Does not require routine coagulation monitoring**
- **Higher risk of bleed if age >75 or weight <50 kg**
- **Can be used in patients with current or history of HIT due to lack of immune-mediated effect on platelets; must still be used with caution**
- **Incredibly long half-life (17-21 hours, prolonged in renal impairment and the elderly) making therapeutic anticoagulation difficult**

PHARMACOLOGICAL MEASURES - DIRECT FACTOR XA INHIBITORS

Drug	Safety	D/I
Rivaroxaban Apixaban Edoxaban	Bleeding	CYP3A4 + PGP inhibitors (e.g. ketoconazole) CYP3A4 + PGP inducers (e.g. St. John's Wort, rifampin, phenytoin)



Reversal agent:
andexanet alfa
(not officially indicated for edoxaban)

COMMENTS

Rivaroxaban (Xarelto®)

- 15 mg BID PO × 3 weeks, then 20 mg PO daily
- **CrCl 15-30 mL/min: use with caution**
- **CrCl <15 mL/min: Avoid use**
- Beyond 6 months, may consider reducing dose to 10 mg PO daily
- **15 mg and 20 mg tablets - must take with food**
- Onset: 1-3 hours, Half-life: 7-11 hours

Apixaban (Eliquis®)

- 10 mg BID PO × 7 days, then 5 mg BID PO for 3-6 months
- Beyond 6 months, may consider reducing dose to 2.5 mg PO BID
- **CrCl 15-29 mL/min: use with caution**
- **CrCl <15 mL/min or undergoing dialysis: Avoid use**
- TIP: **apixaban preferred** > dabigatran, rivaroxaban, edoxaban **in patients with history of GI bleeding**
- Onset: 1-3 hours, half-life: 8-12 hours

Edoxaban (Lixiana®)

- 5-10 days initial Tx with parenteral anticoagulant (usually LMWH)
- 60 mg PO daily OR
- 30 mg PO daily if CrCl 15-50 mL/min, ≤60 kg, or PGP inhibitors (except amiodarone and verapamil)
- **CrCl <15 mL/min or undergoing dialysis: Avoid use**
- Onset: 1-3 hours, half-life: 10-14 hours

PHARMACOLOGICAL MEASURES - DIRECT THROMBIN INHIBITORS

Drug	MOA	Safety	D/I
Dabigatran (Pradaxa®)	Direct thrombin inhibitor (clotting factor IIA)	• Bleeding • GI upset (dyspepsia) • Case reports of ↑MI risk (coronary artery events appear to occur more often with dabigatran than with VKA)	• Anti-platelet & anticoagulants • PGP inducers & inhibitors • Antacids – separate 2 hours

COMMENTS

- 5-10 days initial Tx with parenteral anticoagulant (usually LMWH)
- Onset: 1-3 hours, Half-life: 7-17 hours
- **Contraindicated when CrCl <30 mL/min**
- VTE treatment dose: 150 mg PO BID
- Lower dose may be considered in patients ≥80 y: 110 mg PO BID (**not studied in VTE Tx**)
- Elderly patients over 75 years old with ≥ 1 risk factor(s) for bleeding, consider 110 mg BID
- Do not break open capsule, crush or expose to moisture, keep in original packaging
- **Reversal agent: idarucizumab**

SWITCHING BETWEEN ANTICOAGULANT THERAPIES

Warfarin to DOAC	Warfarin to apixaban or dabigatran	Stop warfarin, start DOAC when INR <2.0
	Warfarin to edoxaban or rivaroxaban	Stop warfarin, start DOAC when INR ≤2.5
UFH to DOAC		Stop UFH infusion, start DOAC immediately (except edoxaban – start 4 hours later)
LMWH to DOAC		Start DOAC at time of next scheduled dose (rivaroxaban/dabigatran – 2 hours before)
DOAC to LMWH/UFH		Start LMWH/UFH at time of next scheduled dose (exception: 12 hours after last dabigatran dose)
DOAC to warfarin		Controversy with overlap vs. bridging

VTE PROPHYLAXIS

Non-pharmacological Measures

- Graduated compression stockings (GCS)
- Intermittent pneumatic compression (IPC) devices
- In the absence of contraindications, DVT prophylaxis should be provided for patients based on assessment of risks
- Duration of treatment depends on the surgery and medical condition
- Recommend early mobilization in all patients

Mechanical options (e.g. GCS, IPC) may be considered instead of pharmacologic if patients are at high-risk of bleeding

Pharmacological Measures

Orthopedic surgeries (hip/knee replacement)	Non-orthopedic surgeries
LMWH, DOACs (rivaroxaban, apixaban, dabigatran), fondaparinux	LMWH, UFH
ASA 81 mg - can be considered for ongoing anticoagulation after initial treatment with rivaroxaban 10 mg daily for 5 days post-op	Less evidence for DOACs
Continue anticoagulant for 14-35 days post-op	Continue anticoagulant at least until discharge

Anticoagulant	Dose for VTE prophylaxis	Dose for VTE Treatment
Apixaban	2.5 mg BID	10 mg BID x 7 days, then 5 mg BID
Rivaroxaban	10 mg once daily	15 mg BID x weeks, then 20 mg daily
Dabigatran	220 mg once daily	150 mg BID



Pharmacological choices are more effective

VTE PROPHYLAXIS

Thromboprophylaxis for hospitalized medical patients

Risk factors	
High-risk patients: Stroke, active cancer, ICU admission	Other risk factors: Age >60 years, reduced mobility, history of CHF, previous VTE, known thrombophilia, recent fracture (within the past 3 months), sepsis, acute inflammatory conditions and acute infectious diseases (e.g. COVID-19)

Duration of thromboprophylaxis: Generally continued until acute care hospital discharge (not usually extended beyond hospitalization)

Usual doses recommended for thromboprophylaxis:

- Dalteparin 5,000 units SC daily
- Enoxaparin 40 mg SC daily
- Tinzaparin 4,500 units SC daily
- Fondaparinux 2.5 mg SC daily
- UFH 5,000 units SC BID may also be considered*

*LMWH is preferred due to its less frequent dosing and lower risk of HIT

Thromboprophylaxis Contraindications

STRONG CONTRAINDICATIONS	RELATIVE CONTRAINDICATIONS
High-risk bleeding features include: - Clinically important bleeding (intracranial hemorrhage) - Severe thrombocytopenia - Untreated major bleeding disorders	- Recent intracranial hemorrhage - Recent paraspinal bleeding - Recent high bleeding risk surgery

SPECIAL POPULATIONS - PREGNANCY

Risks of VTE	Risk of VTE increases 5- to 10-fold during pregnancy and 15- to 35-fold after delivery compared to non-pregnant women of the same age
Monitoring for VTE	Monitor for signs of VTE throughout pregnancy and act on clinical suspicions similar to non-pregnancy diagnosis
Treatment of VTE	DOC: LMWH; treatment suggested to continue throughout pregnancy and for at least 6 weeks postpartum (minimum total duration of 3 months)

- **Avoid: warfarin and DOACs**
- **Safest agents:** UFH and LMWH (LMWH does not cross the placenta and increased risks of fetal bleeding/teratogenic events have not been reported)
- **Breastfeeding/postpartum:** **avoid DOACs**, safe options = LMWH, UFH, warfarin

MONITORING PARAMETERS

Efficacy Endpoints

Parameter	Degree of Change	Timeframe
Signs and symptoms of DVT/PE: pain, swelling, SOB	Improved	2-5 days

Safety Endpoints

Parameter	Degree of Change	Timeframe
Future VTE	Prevented	Duration of therapy
INR	Target for INR of 2.0-3.0	Duration of Warfarin therapy
aPTT	Within target (depends on agent)	Duration of Heparin Therapy
CBC, Platelets, SCr	Within normal range	Duration of therapy
Excessive bleeding/AE from anticoagulant	Prevented	Duration of therapy

CASE 1

AD is a 50-year-old male who presents to the ER with unilateral swelling of his left lower leg. He complains of pain and the lower limb is quite red. AD explains that he was hospitalized 2 months ago for approximately two days after a total knee replacement. He has been quite sedentary since. AD tells the staff he was on a “blood thinner” after his surgery but only for the first 15 days post-operation. That finished about 6 weeks ago. AD has a past medical history significant for type 2 diabetes mellitus, hypertension, and dyslipidemia. He currently takes metformin 1000 mg PO BID, gliclazide SR 60 mg PO daily, ramipril 10 mg PO daily, rosuvastatin 20 mg PO QHS, omega-3 1 capsule PO daily, and vitamin E 400 units PO daily. He drinks 1-2 alcoholic drinks per week. He has no allergies to medications.

1. AD is diagnosed with a DVT. Which of the following would NOT be an initial treatment option?
 - a) Rivaroxaban
 - b) Fondaparinux
 - c) Enoxaparin
 - d) Warfarin



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2. What is the most appropriate duration of therapy to recommend?
- a) 1 month
 - b) 3 months
 - c) 8 months
 - d) 1 year



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3. If AD had presented with active stage 1 melanoma, what would be the appropriate duration of therapy?
- a) 3 months
 - b) 6 months
 - c) Until the malignancy resolves
 - d) 9 months



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4. If warfarin was chosen as the oral anticoagulant for maintenance, what would the target INR be if AD had a mechanical mitral valve?
- a) 2.0-3.0
 - b) 1.0-2.0
 - c) 2.5-3.5
 - d) 1.5-2.5



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5. If AD's CrCl was 45 mL/min at the time of his surgery, which of the following would be a safe, effective, and convenient option to use for 15 days of DVT prophylaxis post-op?
- a) Warfarin
 - b) Rivaroxaban
 - c) Fondaparinux
 - d) UFH



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CHANGE LOG

July 2025

- Formatting changes made throughout; legend and reference list updated
- Few slides rearranged for better flow, and wording updated to more closely match guidelines
- Slide 4: Added fibrin point under hypercoagulable state definition
- Slide 7: Minor additions made for clarification
- Slide 8: Added thalidomide and bevacizumab to chemotherapy list, replaced L-asparaginase with pegaspargase
- Slide 10: Added footnotes
- Slide 11: Edited point #1
- Slide 14: Added information to 'ambulation'
- Slide 15: Modified wording in brackets for fondaparinux for clarity
- Slide 24: Updated table

September 2025

- Formatting changes made throughout, some slides re-arranged
- Slide 18: Added information to drug charts
- Slide 22: Added information to drug charts
- Slide 23: Added PK information
- Slide 26: New

February 2026

- Slide 2: Added new NAPRA competencies
- Slide 10: changed "limited data" to "off-label"
- Slide 16: Added Minimum 3 months to proximal DVT as per thrombosis Canada
- Slide 18: Added LMWH dose adjustment specific
- Slide 22: Added that andexanet alpha is not officially indicated for edoxaban